

Julian Andres Banguera Torres

(910) 929-0732 | julian-bt@outlook.com | <https://www.linkedin.com/in/julian-torres-9485161b0/>

EDUCATION

- **University of North Carolina at Charlotte** 08/2021 – 05/2023
Master of Science in Mathematical Finance | GPA: 3.60
- **University of North Carolina at Charlotte** 07/2020 – 12/2022
Bachelor of Science in Systems Engineering, Minors in Math, Statistics and Data Science | GPA: 3.94
- **Central Carolina Community College** 01/2017 – 05/2020
Associate in engineering | GPA: 3.8

RELEVANT COLLEGE PROJECTS & CERTIFICATIONS

- Developed an econometric model in Alteryx predicting loan default probability within 4 years of origination, using it end to end for ingestion, cleaning and comparing logistic, boosting, and bagging algorithms.
- Databricks Certified Data Engineer Associate certification

SKILLS

- Data Pipelines(SQL , Databricks, Fivetran, MSSQL), Python (Pandas, PySpark, Airflow, Dash), Tableau, Alteryx, Figma, Excel, R, Java, Copilot, Jira

WORK EXPERIENCE

Technology Associate

08/2023 – 05/2026

Morgan Stanley, Atlanta, GA

- Replaced a naive forecasting approach that assumed future revenue would mirror the trailing six months by developing an ARIMA model. Airflow orchestrated Fivetran ingestion from operational data sources, followed by a Databricks ETL pipeline that transformed and validated the data, then trained ARIMA models in parallel to deliver financial advisor level revenue forecasts.
- Built end to end Tableau dashboard and pipeline to drive and track data type change adoption:
 - Performed ad hoc SQL analysis on lineage and consumption data to determine the best replacement data type for a key attribute, then used the Figma to Tableau plugin to prototype layouts for early stakeholder validation.
 - Implemented a pre built Tableau extension enabling users to submit bulk update changes directly from the dashboard, writing to a dedicated Databricks table.
 - Used Databricks Auto Loader and Apply Changes Into to apply SCD Type 2 modeling on incoming updates, preserving full change history.
 - Computed completion KPIs on top of the SCD Type 2 history to track new data type adoption progress, precomputing them in the gold layer for faster dashboard performance.
 - Orchestrated two independent pipelines, one for scheduled ingestion and transformation, the other streaming bulk update changes.
 - Improved Spark performance using liquid clustering, predictive optimization, and adaptive query execution to handle data skew.
 - Implemented Tableau row level security using Virtual Connection Data Policies, matching authenticated users against application entitlements to ensure users could update only the applications they belonged to.
- Built a Tableau workbook and automated pipeline to replace monthly manual excel reporting, orchestrating Fivetran ingestion, a Databricks ETL pipeline, and Tableau extract refreshes with Airflow operators interacting with APIs across all three. The ETL pipeline decoupled legacy SQL Server stored procedures into a medallion architecture.

- Reduced tabular Tableau dashboard load time by 85% by redesigning it in Python Dash with AG Grid, using server side infinite scrolling and on demand data retrieval via Dash callbacks instead of full dataset loads into memory.

Technology Summer Analyst intern

06/2022 – 08/2022

Morgan Stanley, Atlanta, GA

- Contributed to the E*Trade acquisition integration by building a PySpark notebook in Databricks that demonstrated a faster alternative to the existing SQL heavy audit process, reading two versions of tables and using hashing to efficiently detect changes between them, reducing run time by 75%.

Technology Summer Analyst intern

06/2021 – 08/2021

Deutsche Bank, Raleigh, NC

- Built a fully interactive Tetris game in Java as a team internal project. Used object-oriented design to model game pieces, board state, and gameplay interactions, and implemented core features such as movement, rotation, line clearing, scoring, and increasing difficulty. Delivered a polished internal game aligned with the bank's engineering standards, tooling and development practices.